KASIN ITO-SATO

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Mechanical Engineer (BS, May 2027) | Mechanical Design | Thermal & Flow Analysis | Assembly & Test | GD&T

EDUCATION

Loyola Marymount University **May 2027**

Bachelor of Science (BS), Mechanical Engineering

Relevant Coursework: Machine Design, Design and Manufacturing, Thermodynamics, Heat Transfer, Fluid Mechanics

ENGINEERING PROJECTS

2.0x Pantograph | Design and Manufacturing, LMU Fall 2024

- **1.98x to 2.00x measured accuracy** on an aluminum linkage we designed in SolidWorks and machined on a mill, bandsaw, and 3D printer, including a replaceable marker holder and an optical breadboard mounting interface.
- **Held true position at MMC for floating 1/4-20 fasteners** through a T = H − F tolerance analysis and ASME Y14.5 drawings, locating holes with the mill's digital readout so as-built parts matched CAD.

Rope-Climbing Robot | Machine Design, LMU Spring 2026

- **Crossed a 38 ft outdoor rope span in 5.5 minutes** (86% of predicted speed) after building, assembling, and testing the motor-driven 3-stage spur gear drivetrain (64T/32T/16T) on a four-person team.
- **Rebuilt the mechanism as a constrained screw slider** after lab testing ruled out the arm design, troubleshooting hooks, slider, and gear engagement through fast 3D-printed and wood part revisions.

Thermal & Flow Analysis | Heat Transfer, LMU Spring 2026

- **Cut required pipe size from NPS 4 to NPS 2** while meeting a 0.8 in/hr melt target, by modeling pipe pressure loss, pump power, convection, and soil conduction for a 1,500 sq ft hydronic loop in MATLAB.
- **Cut a heated block from 137.5 to 89.0°C with a fin array** and measured convection coefficients of 12.8 W/m²K (free) and 48.3 W/m²K (forced) using Arduino thermocouple data acquisition.
- **Measured forced-convection heat transfer on a heated cylinder at 3 air speeds (4.3 to 6.2 m/s)**, confirming it rose with Reynolds number (6,400 to 9,200) as flow theory predicts.

Teardown, Part Sourcing & Material Selection | Machine Design and Materials Labs, LMU 2025 - 2026

- **Disassembled, inspected, and reassembled a power drill**, measuring a press-fit bearing in place with calipers and sourcing a McMaster-Carr replacement part.
- **Ranked the top 10 lightweight materials for buckling, bending, and torsion** using material indices and shape factors in CES EduPack.
- **Validated 7050-T7 aluminum yield within about 2% of published values** (ASTM E8, Instron 4505) and documented the method and results in a formal engineering test report.

PROFESSIONAL EXPERIENCE

WSP in the US **May 2025 - January 2026**

Mechanical Engineering Intern *Honolulu, HI*

- **Sized mechanical equipment from loads cross-checked in 2 models** (IES-VE, TRACE 700), calculated airflow pressure losses, and coordinated equipment selection with manufacturers by reviewing their submittals against specifications.
- **Built 3 engineering analysis tools in Excel that the team adopted** for pressurization, ventilation, and exhaust flow, on my own initiative, with step-by-step guides for each.
- **Revised mechanical layouts and drawings in Revit and AutoCAD** through multiple design stages, and troubleshot a drawing vs. installation discrepancy on site with the install team.

SKILLS

Mechanical Design: SolidWorks (parts, assemblies, drawings), Onshape, AutoCAD, Revit, GD&T (ASME Y14.5), tolerancing, DFM
Thermal & Fluids: Heat transfer, flow analysis, pipe pressure loss, pump power, thermocouple data acquisition, MATLAB/Simulink
Materials & Test: CES EduPack material selection, tensile and fatigue testing, metallography, engineering test reports
Fabrication: Mill, lathe, CNC, waterjet, laser cutting, 3D printing, welding, mechanical assembly and disassembly
Certifications: FE Mechanical exam scheduled November 2026; Rhinoceros CAD Level 2 **Languages:** English, Japanese